

DATA 260 - Agentic AI

Homework 6 - Evidence Report

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SID4: 7085

Repository: data260-7085

Branch: hw6

Configuration: PORT_BASE=8785, PREFIX=s7085, SEED=7085, VERIFY_SEED=267085, DOMAIN_ID=5

Local model: Ollama llama3.2:3b

Repository: <https://github.com/shoddy16/data260-7085>

Submission status

This report was generated from the current HW6 branch and checked-in machine-readable results. Live MongoDB/Ollama/API screenshots are not claimed unless captured from an actual running service. The current environment could not install the missing FastAPI/PyMongo dependencies because the package proxy returned a corrupted download.

Part 1 - MongoDB Document API

The implementation adds a MongoDB-backed document view at `/api/documents` with required short text, optional description, constrained status/category fields, an inspection datetime, automatic timestamps, Pydantic validation, and MongoDB JSON Schema validation. CRUD handlers return explicit validation, not-found, database, and success status codes.

Relevant implementation: `hw6_mongo.py`, `hw6_memory.py`, `routers/hw6.py`, and `main.py`. Live MongoDB screenshots remain pending because MongoDB and PyMongo were unavailable in this run.

Part 2 - Short-, Long-, and Episodic Memory

The `/api/chat`, `/api/memory/{user_id}`, and `/api/aggregate/{user_id}` routes use messages, summaries, and episodes. Chat turns save user and assistant messages, assemble system/session/lifetime/short-term/episodic context, create deterministic local embeddings, retrieve episodes by cosine similarity, and trigger summaries every configured number of user messages.

Offline deterministic tests verify the sliding window, episodic retrieval, prompt assembly, and summary trigger. A live conversation screenshot and MongoDB Compass evidence require the local services and dependencies.

Part 3 - What Memory Costs

Setting	Mean prompt tokens/turn	Mean latency (ms)	Probe quality
SHORT_TERM_N=4	299.733	0.007	3.5/5
SHORT_TERM_N=12	412.2	0.002	4.0/5
SHORT_TERM_N=30	601.267	0.007	4.0/5
compression_policy	296.733	0.007	3.9/5

The 30-turn deterministic sweep selected `compression_policy` as the best measured token-cost setting and `SHORT_TERM_N=30` as the worst. Estimated cost for 10,000 conversations was \$119.0199 best and \$210.3801 worst using the required input/output prices.

Raw records: `reports/hw06/raw/memory_token_records.csv`, `memory_cost_comparison.json`, `memory_cost_summary.json`.
Probe definitions: `memory_probes.yaml`.

Part 4 - File-Based Memory and Consolidation

`AGENT.md` now contains project, technology, coding, testing, and prohibited-action instructions. `MEMORY.md` stores durable facts and includes a timestamp. The `/memory`, `/compact`, `/dream`, and `/help` commands are implemented in `file_memory.py`. The original contradictory 55-line fixture is preserved in `reports/hw06/raw/MEMORY_original.md`.

The actual consolidation reduced 56 non-empty/comment-filtered source lines to 32 output facts, merged duplicates, and explicitly recorded the MySQL/MongoDB contradiction for review. Deterministic tests verify command output and contradiction handling.

Verification and AI Use

Verification output is written by `verify_hw6.py` to `reports/hw06/verification.json`. `AI_USE.md` documents the assistant's role, independent verification, detected problems, and corrective changes. `RUN_LOG.txt` records the implementation and experiment commands.

Required final GitHub push/tag and live-service screenshots are not reported as complete until the missing dependency installation and service checks succeed.